

Approximation and Taylor's Theorem

Engineering Mathematics–I (CSE-1001)

BTech 1st Year, IIIT Bhopal

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To the Student

This note explains **approximation** in simple words: how a complicated function can be replaced, near a chosen point, by a straight line, then by a parabola, then by a polynomial of any degree we please — and, most importantly, how to say *how wrong* the replacement is. It is written for students meeting the topic properly for the first time. Every idea is built slowly: first we look at a real situation where the idea is needed, then we look at a picture, then we learn the definition, and only then do we work through examples.

This is the first note of the course set, and it stands on its own. You do not need the companion notes on *Sequences* and *Series* to read Sections 1–6: all you need is differentiation. Only the last part, where the polynomials are allowed to run on forever and become a *Taylor series*, uses ideas from those notes, and Section 7 says clearly where.

Read the sections in order. Everything here rests on one formula — the Taylor polynomial of Section 5 — and everything before it is the special case $n = 1$ and $n = 2$ of that formula. Keep pen and paper beside you: compute each approximation yourself and compare it against a calculator, because the whole subject is about the size of the gap between the two.

How to use these notes:

1. Attend the class lecture for each topic.
2. Read the corresponding section of these notes carefully, working through each example yourself.
3. Stop at every *Try it yourself* box and attempt it on the spot. There are seven of them, and each takes only two or three minutes.
4. Once you have finished a section, attempt the matching questions from the companion question bank (*Approximation and Taylor's Theorem — Question Bank*) for self-assessment.

Good luck, and enjoy the topic.

Notation. Throughout, f' , f'' , f''' and $f^{(n)}$ denote successive derivatives, $\mathbb{N} = \{1, 2, 3, \dots\}$, and a (or x_0) denotes the **base point** about which we approximate. We write $n! = 1 \cdot 2 \cdot 3 \cdots n$ and $0! = 1$.

1 Why Approximate?

Most functions an engineer meets cannot be evaluated exactly. There is no finite recipe for $\sqrt{4.05}$, for $\sin(0.02)$, for $e^{0.01}$. What there *is* — and all that a computer ever has — is addition, subtraction, multiplication and division. So every such value must be reached by a polynomial, because polynomials are exactly the functions those four operations can build.

Four situations, one piece of mathematics

1. Your calculator finding $\sqrt{4.05}$. It has no square-root circuit. It knows $\sqrt{4} = 2$ exactly, and it uses the *slope* of $\sqrt{x}$ at 4 to step across the remaining 0.05. Two questions follow at once: how close does that get, and how much closer does a better polynomial get?

2. GPS satellites. A satellite moving at speed v runs its clock slow relative to yours by a factor

$$\left(1 - \frac{v^2}{c^2}\right)^{-1/2}.$$

With $v = 4$ km/s and $c = 3 \times 10^5$ km/s, $v^2/c^2 \approx 10^{-10}$ — far too small for a calculator to handle gracefully, but exactly the regime where $(1 + u)^r \approx 1 + ru$ is superb. Engineers use this approximation to offset the satellites' transmitter frequencies. Your phone's position depends on it.

3. The pendulum. The exact equation of a swinging pendulum, $\ddot{\theta} = -\frac{g}{L} \sin \theta$, cannot be solved in elementary functions. Replace $\sin \theta$ by θ — valid for small swings — and it becomes the simple harmonic oscillator every physics course solves in a line. The entire theory of small oscillations rests on one linear approximation.

4. Linearising a control system. Any nonlinear system — a robot arm, a chemical reactor, an aircraft — is designed by replacing its behaviour near an operating point by a linear model. That model *is* the linear approximation of this chapter, in several variables.

Zoom in and everything is straight

Take any smooth curve and magnify a small piece of it. The more you magnify, the straighter it looks — which is precisely what it means for f to be differentiable at a . Linear approximation is nothing more than taking that observation seriously and using the straight line instead of the curve.

Quadratic approximation is the next step: after the line, use the parabola that also curves the right way. Taylor's Theorem is the general step: use the polynomial of degree n that matches f in value and in the first n derivatives.

2 Linear Approximation

Definition 2.1 (Linear approximation). Let f be differentiable on an interval containing the point a . The **linear approximation** (or **tangent line approximation**) to f at a is the polynomial

$$P_1(x) = L(x) = f(a) + f'(a)(x - a)$$

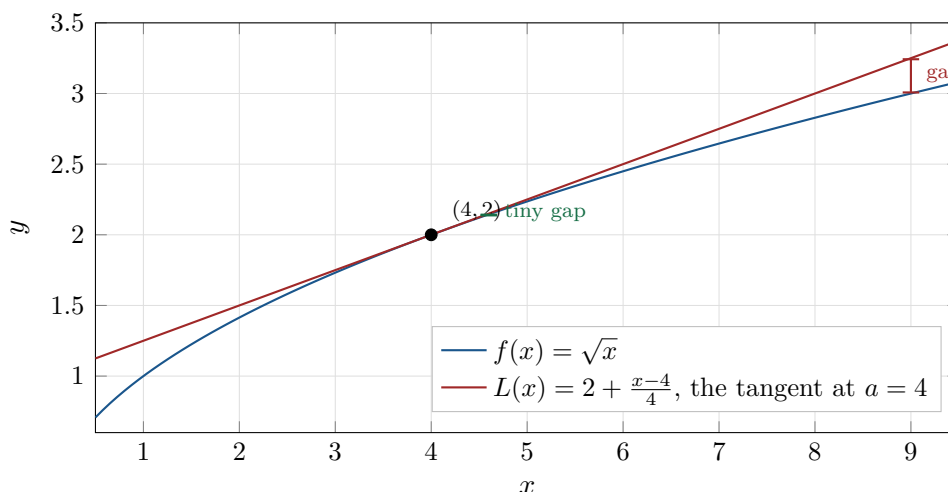


Figure 1: The tangent line at $a = 4$ is indistinguishable from $\sqrt{x}$ close to $x = 4$, and visibly wrong by $x = 9$. This is the whole subject in one picture: a linear approximation is excellent *near the base point* and deteriorates as you move away. The rest of this note makes “excellent” and “deteriorates” into numbers.

and we write $f(x) \approx f(a) + f'(a)(x - a)$ for x near a .

P_1 is the unique polynomial of degree ≤ 1 with

$$P_1(a) = f(a) \quad \text{and} \quad P_1'(a) = f'(a),$$

that is, the unique line through $(a, f(a))$ with the same slope as f there. Geometrically it is the tangent line of Figure 1.

Example 2.2. Compute $\sqrt{4.05}$ by linear approximation.

Solution. Take $f(x) = \sqrt{x}$ and $a = 4$, the nearest point at which the square root is easy.

$$f(4) = 2, \quad f'(x) = \frac{1}{2\sqrt{x}} \implies f'(4) = \frac{1}{4}.$$

Hence

$$L(x) = 2 + \frac{1}{4}(x - 4), \quad \text{so} \quad \sqrt{4.05} \approx 2 + \frac{1}{4}(0.05) = 2.0125.$$

The true value is 2.01246118, so the error is 0.0000388 — about four parts in a hundred thousand, from one multiplication. $\square$

Choose the base point, and choose it well

Two things decide the quality of a linear approximation: how close x is to a , and how sharply f curves in between. The base point must be a place where $f(a)$ and $f'(a)$ are *known exactly* — that is the whole point of choosing $a = 4$ rather than $a = 4.05$. Choosing $a = 0$ out of habit when the question is about $x = 4.05$ is a guaranteed way to get nonsense.

2.1 The basic list

With base point $a = 0$ and $|x| \ll 1$, the five approximations below cover most of what is ever needed. They are worth memorising.

	Approximation	$f(0)$	$f'(0)$
1	$\sin x \approx x$	0	1
2	$\cos x \approx 1$	1	0
3	$e^x \approx 1 + x$	1	1
4	$\ln(1 + x) \approx x$	0	1
5	$(1 + x)^r \approx 1 + rx$	1	r

Proofs. Each is $L(x) = f(0) + f'(0)x$ with the values in the table.

1. $f(x) = \sin x$, $f'(x) = \cos x$, so $f(0) = 0$, $f'(0) = 1$ and $L(x) = 0 + 1 \cdot x = x$.
2. $f(x) = \cos x$, $f'(x) = -\sin x$, so $f(0) = 1$, $f'(0) = 0$ and $L(x) = 1$.
3. $f(x) = e^x$, $f'(x) = e^x$, so $f(0) = f'(0) = 1$ and $L(x) = 1 + x$.
4. $f(x) = \ln(1 + x)$, $f'(x) = \frac{1}{1+x}$, so $f(0) = 0$, $f'(0) = 1$ and $L(x) = x$.
5. $f(x) = (1 + x)^r$, $f'(x) = r(1 + x)^{r-1}$, so $f(0) = 1$, $f'(0) = r$ and $L(x) = 1 + rx$. $\square$

2.2 Changing the base point

A base point other than 0 can always be moved to 0 by a substitution, and doing so lets the standard list be used.

Example 2.3. Find the linear approximation of $f(x) = \ln x$ near $x = 1$.

Solution. Here $f(1) = \ln 1 = 0$ and $f'(x) = \frac{1}{x}$, so $f'(1) = 1$. Hence

$$\ln x \approx 0 + 1 \cdot (x - 1) = x - 1.$$

Now substitute $u = x - 1$, so that $x = 1 + u$ and $u \approx 0$:

$$\ln(1 + u) \approx u,$$

which is entry 4 of the basic list. The two statements are the same fact, written about two different base points. $\square$

2.3 Combining approximations algebraically

Rather than differentiate a complicated f , it is usually faster to break it into pieces, approximate each piece from the list, multiply out, and **discard every power of x above the first**.

Example 2.4. Find the linear approximation of $f(x) = \frac{e^{-2x}}{\sqrt{1+x}}$ near $x = 0$.

Solution. Approximate the two factors separately.

$$e^{-2x} \approx 1 + (-2x) = 1 - 2x \quad (\text{entry 3 with } u = -2x),$$

$$\frac{1}{\sqrt{1+x}} = (1+x)^{-1/2} \approx 1 - \frac{1}{2}x \quad (\text{entry 5 with } r = -\frac{1}{2}).$$

Multiply:

$$f(x) \approx (1 - 2x) \left(1 - \frac{1}{2}x\right) = 1 - \frac{1}{2}x - 2x + x^2 = 1 - \frac{5}{2}x + x^2.$$

Discard the x^2 term — we have already thrown away x^2 and higher terms in forming each factor, so keeping this one would be dishonest. Hence

$$\frac{e^{-2x}}{\sqrt{1+x}} \approx 1 - \frac{5}{2}x$$

As a bonus, comparing with $L(x) = f(0) + f'(0)x$ we can read off $f(0) = 1$ and $f'(0) = -\frac{5}{2}$ without ever differentiating f . $\square$

Discard consistently

When multiplying two linear approximations, an x^2 term always appears. It must be thrown away. Its coefficient is not trustworthy, because the x^2 terms that were already discarded from each factor would have contributed to it as well. Keep terms only up to the order you are working to — first order here, second order in Section 3.

The GPS calculation

Recall the time-dilation factor $\left(1 - \frac{v^2}{c^2}\right)^{-1/2}$. Apply entry 5 with $u = -\frac{v^2}{c^2}$ and $r = -\frac{1}{2}$:

$$\left(1 - \frac{v^2}{c^2}\right)^{-1/2} \approx 1 + \left(-\frac{1}{2}\right) \left(-\frac{v^2}{c^2}\right) = 1 + \frac{1}{2} \frac{v^2}{c^2}.$$

With $v = 4$ km/s and $c = 3 \times 10^5$ km/s we get $\frac{v^2}{c^2} \approx 1.78 \times 10^{-10}$, so the satellite clock differs from yours by about one part in 10^{10} — roughly 8 microseconds a day. Light travels 2.4 km in 8 microseconds, so uncorrected GPS would put you in the wrong city by evening. The correction is applied using exactly the approximation above.

Try it yourself

Find the linear approximation near $x = 0$ of

$$(i) \sqrt[4]{1+x} \quad (ii) \frac{1}{1-3x} \quad (iii) e^x \cos x$$

Answers: $1 + \frac{x}{4}$; $1 + 3x$; $1 + x$.

3 Quadratic Approximation

A straight line cannot bend. If f curves appreciably over the interval of interest, the next polynomial up will do much better.

Definition 3.1 (Quadratic approximation). Let f be twice differentiable near a . The **quadratic approximation** to f at a is

$$P_2(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2$$

It is the unique polynomial of degree ≤ 2 with

$$P_2(a) = f(a), \quad P'_2(a) = f'(a), \quad P''_2(a) = f''(a),$$

so it matches f in position, slope *and* curvature.

Where the $\frac{1}{2}$ comes from

The $\frac{1}{2}$ is exactly what makes the formula *exact* on quadratics. Take $f(x) = a_0 + a_1x + a_2x^2$ and base point 0. Then

$$f(0) = a_0, \quad f'(0) = a_1, \quad f''(0) = 2a_2,$$

so $a_2 = \frac{f''(0)}{2}$. Feeding these back gives $P_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 = f(x)$. Without the $\frac{1}{2}$ the formula would fail on the very functions it is built from. The same reasoning produces the $\frac{1}{n!}$ in Taylor's Theorem.

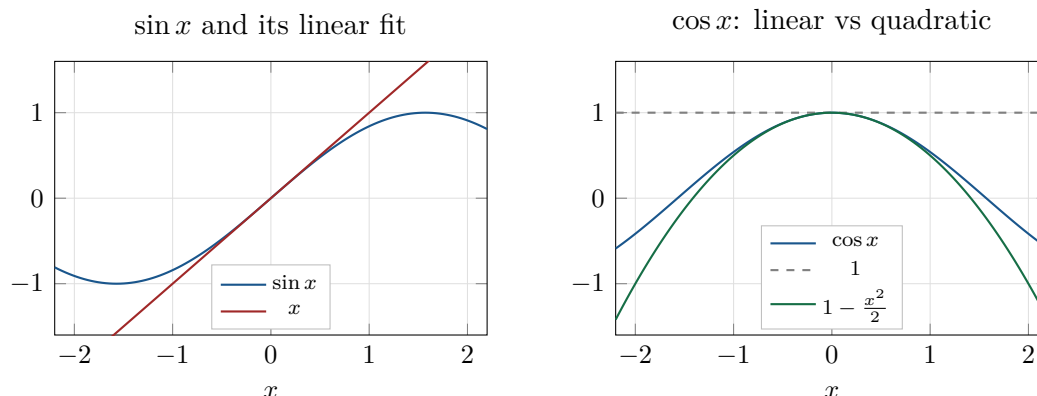


Figure 2: On the left, $\sin x \approx x$ is excellent near 0 and fails by $x \approx 1$. On the right, the linear approximation $\cos x \approx 1$ is a flat line that cannot follow the curve at all, while the quadratic $1 - \frac{x^2}{2}$ tracks it closely out to $|x| \approx 1$. When $f'(a) = 0$, the quadratic term is doing *all* the work.

3.1 The basic list, to second order

Quadratic approximation near $x = 0$

- 1 $\sin x \approx x$
 - 2 $\cos x \approx 1 - \frac{x^2}{2}$
 - 3 $e^x \approx 1 + x + \frac{x^2}{2}$
 - 4 $\ln(1+x) \approx x - \frac{x^2}{2}$
 - 5 $(1+x)^r \approx 1 + rx + \frac{r(r-1)}{2}x^2$
-

Notice entry 1: the quadratic approximation of $\sin x$ is still x , because $f''(0) = -\sin 0 = 0$. The first correction to $\sin x$ is cubic, which is why $\sin x \approx x$ is so unusually accurate.

Proof of entry 4. $f(x) = \ln(1+x)$ gives $f(0) = 0$; $f'(x) = \frac{1}{1+x}$ gives $f'(0) = 1$; $f''(x) = -\frac{1}{(1+x)^2}$ gives $f''(0) = -1$. Hence

$$P_2(x) = 0 + 1 \cdot x + \frac{-1}{2}x^2 = x - \frac{x^2}{2}. \quad \square$$

Example 3.2. Compute $\sqrt{4.05}$ by quadratic approximation.

Solution. With $f(x) = \sqrt{x}$ and $a = 4$:

$$f(4) = 2, \quad f'(4) = \frac{1}{4}, \quad f''(x) = -\frac{1}{4}x^{-3/2} \implies f''(4) = -\frac{1}{32}.$$

Hence

$$P_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2,$$

and at $x = 4.05$,

$$\sqrt{4.05} \approx 2 + \frac{1}{4}(0.05) - \frac{1}{64}(0.05)^2 = 2 + 0.0125 - 0.0000390625 = 2.0124609375. \quad \square$$

Method	Value	Error
Linear $P_1(4.05)$	2.012500000	3.9×10^{-5}
Quadratic $P_2(4.05)$	2.012460938	2.4×10^{-7}
True value	2.012461180	—

One extra term has improved the accuracy by a factor of about 160. That gain is what Section 4 quantifies in general.

3.2 Combining to second order

The algebraic trick of Section 2 works here too; simply keep terms up to x^2 instead of up to x .

Example 3.3. Find the quadratic approximation of $f(x) = \frac{e^{-2x}}{\sqrt{1+x}}$ near $x = 0$.

Solution. Using entries 3 and 5 to second order,

$$e^{-2x} \approx 1 - 2x + \frac{(-2x)^2}{2} = 1 - 2x + 2x^2,$$

$$(1+x)^{-1/2} \approx 1 - \frac{1}{2}x + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2}x^2 = 1 - \frac{1}{2}x + \frac{3}{8}x^2.$$

Multiply, keeping terms up to x^2 :

$$f(x) \approx \left(1 - 2x + 2x^2\right) \left(1 - \frac{1}{2}x + \frac{3}{8}x^2\right) = 1 - \frac{5}{2}x + \left(2 + 1 + \frac{3}{8}\right)x^2 = 1 - \frac{5}{2}x + \frac{27}{8}x^2.$$

(The x^2 coefficient collects 2, from $2x^2 \cdot 1$; 1, from $(-2x)(-\frac{1}{2}x)$; and $\frac{3}{8}$, from $1 \cdot \frac{3}{8}x^2$.) Terms in x^3 and above are discarded. $\square$

Try it yourself

Find the quadratic approximation near $x = 0$ of

$$(i) e^x \sin x \quad (ii) \sqrt{1+x} \ln(1+x) \quad (iii) \frac{\cos x}{1+x}$$

Answers: $x+x^2$; x (the x^2 terms cancel — check this, it is a good test of the method); $1-x+\frac{x^2}{2}$.

4 How Good Is the Approximation? Error Estimates

So far we have compared against a calculator. That is not mathematics: it requires knowing the answer in advance. We now bound the error *without* knowing the true value.

Theorem 4.1 (Extended Mean Value Theorem). *Let f' be continuous on $[a, b]$ and let f'' exist on (a, b) . Then for any x in $[a, b]$ with $x \neq a$ there exists c strictly between a and x such that*

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(c)}{2}(x-a)^2.$$

This is the ordinary Mean Value Theorem with one more term, and it is the case $n = 1$ of Taylor's Theorem (Section 5).

Corollary 4.2 (Error in linear approximation). *With $E_1(x) = f(x) - P_1(x)$ and $M = \sup\{|f''(t)| : t \text{ between } a \text{ and } x\} < \infty$,*

$$|E_1(x)| \leq \frac{M}{2} (x-a)^2$$

Corollary 4.3 (Error in quadratic approximation). *With $E_2(x) = f(x) - P_2(x)$ and $M = \sup\{|f'''(t)| : t \text{ between } a \text{ and } x\} < \infty$,*

$$|E_2(x)| \leq \frac{M}{6} |x-a|^3$$

Read the bounds, do not just memorise them

Both bounds have the same shape: *a bound on the next derivative*, divided by a *factorial*, times the *distance to the base point raised to the next power*. Three consequences follow immediately.

- Halving $|x-a|$ divides the linear error by 4 and the quadratic error by 8. Staying near the base point pays, and pays more for higher-degree approximations.
- A function with a large second derivative — one that bends sharply — is badly served by its tangent line. M is the mathematical statement of “how much it bends”.
- The factorial in the denominator is what eventually wins in Taylor's Theorem, because $n!$ outgrows everything.

Example 4.4. Estimate the error when $\sqrt{1.1}$ is computed from the linear and quadratic approximations to $\sqrt{x}$ at $a = 1$.

Solution. Here $f(x) = \sqrt{x}$, $f'(x) = \frac{1}{2}x^{-1/2}$, $f''(x) = -\frac{1}{4}x^{-3/2}$, $f'''(x) = \frac{3}{8}x^{-5/2}$.

Linear. $P_1(x) = 1 + \frac{1}{2}(x-1) = \frac{x}{2} + \frac{1}{2}$, so $P_1(1.1) = 1.05$. On $[1, 1.1]$, $|f''(t)| = \frac{1}{4}t^{-3/2} \leq \frac{1}{4}$, so $M = \frac{1}{4}$ and

$$|E_1(1.1)| \leq \frac{1/4}{2}(0.1)^2 = \frac{0.01}{8} = \frac{1}{800} = 0.00125.$$

The true value is $\sqrt{1.1} = 1.04880$, so the actual error is 0.00120 — comfortably inside the bound, and close to it.

Quadratic. $P_2(x) = 1 + \frac{1}{2}(x-1) - \frac{1}{8}(x-1)^2$, so $P_2(1.1) = 1.05 - 0.00125 = 1.04875$. On $[1, 1.1]$, $|f'''(t)| = \frac{3}{8}t^{-5/2} \leq \frac{3}{8}$, so

$$|E_2(1.1)| \leq \frac{3/8}{6}(0.1)^3 = \frac{3 \times 0.001}{48} = \frac{1}{16000} = 0.0000625.$$

The quadratic value 1.04875 is indeed within 0.00005 of the true 1.04880. $\square$

***M* is a supremum, not a value at a point**

M must bound $|f''|$ (or $|f'''|$) *throughout* the interval between a and x — not merely at a , and not merely at x . Where the derivative is monotone, as in the example above, the supremum is attained at one endpoint and is easy to find; otherwise you must think. Using $|f''(a)|$ in place of M is a common and costly error.

Try it yourself

Use Corollary 4.2 to bound the error in the approximation $e^{0.1} \approx 1.1$, taking $a = 0$.
Hint: $|f''(t)| = e^t \leq e^{0.1} < 1.2$ on $[0, 0.1]$. *Answer:* $|E_1| \leq 0.006$; the true error is 0.00517.

5 Taylor's Theorem

Linear and quadratic are the cases $n = 1$ and $n = 2$ of one general construction.

Definition 5.1 (Taylor polynomial). Let f have n derivatives at a . The **Taylor polynomial of degree n** for f about a is

$$P_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

When $a = 0$ it is also called the **Maclaurin polynomial**.

P_n is the unique polynomial of degree $\leq n$ with

$$P_n(a) = f(a), \quad P'_n(a) = f'(a), \quad \dots, \quad P_n^{(n)}(a) = f^{(n)}(a),$$

so it agrees with f to as high an order as its degree permits.

Theorem 5.2 (Taylor's Theorem, with Lagrange remainder). *Let $f^{(n)}$ be continuous on the closed interval between a and x , and let $f^{(n+1)}$ exist on the open interval between them. Then there exists a point c strictly between a and x such that*

$$f(x) = P_n(x) + R_n(x), \quad R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$

The term $R_n(x)$ is called the **remainder** (or **error term**) of order n .

Taylor's Theorem generalises the Mean Value Theorem

Put $n = 0$. Then $P_0(x) = f(a)$ and the theorem reads

$$f(x) = f(a) + f'(c)(x - a),$$

which is exactly the Mean Value Theorem. Taking $n = 1$ gives the Extended Mean Value Theorem of Section 4. So this one statement contains both, and extends them to every order.

The value of c is *not* known — the theorem only says it exists somewhere between a and x . That is enough, because we never need c itself, only a bound on $f^{(n+1)}$ over the interval it lives in. That is the content of Section 6.

Try it yourself

Write down the Taylor polynomial P_3 for

(i) $f(x) = e^x$ about $a = 0$;

(ii) $f(x) = \frac{1}{x}$ about $a = 1$.

Answers: $1 + x + \frac{x^2}{2} + \frac{x^3}{6}$; and $1 - (x - 1) + (x - 1)^2 - (x - 1)^3$.

Proof of Taylor's Theorem (*)

This proof is included for completeness and may be omitted on a first reading. The idea is to build a function to which Rolle's theorem applies.

Proof. Fix a and x with $x \neq a$. Choose the number M so that

$$f(x) = f(a) + f'(a)(x - a) + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n + \frac{M}{(n + 1)!}(x - a)^{n+1}; \quad (*)$$

this is possible because $(x - a)^{n+1} \neq 0$, so the equation determines M uniquely. The whole task is to show that $M = f^{(n+1)}(c)$ for some c between a and x .

Define, for t between a and x ,

$$g(t) = f(t) + f'(t)(x - t) + \frac{f''(t)}{2!}(x - t)^2 + \cdots + \frac{f^{(n)}(t)}{n!}(x - t)^n + \frac{M}{(n + 1)!}(x - t)^{n+1}.$$

Substituting $t = x$ collapses every term but the first, giving $g(x) = f(x)$; and substituting $t = a$ reproduces the right-hand side of (*), so $g(a) = f(x)$ as well. Hence $g(a) = g(x)$, and g is continuous on the closed interval and differentiable on the open one. By **Rolle's theorem** there exists c strictly between a and x with $g'(c) = 0$.

Now differentiate g . Each product $\frac{f^{(k)}(t)}{k!}(x - t)^k$ contributes, by the product rule,

$$\frac{f^{(k+1)}(t)}{k!}(x - t)^k - \frac{f^{(k)}(t)}{(k - 1)!}(x - t)^{k-1},$$

and consecutive terms cancel in pairs — the sum telescopes. All that survives is

$$g'(t) = \frac{f^{(n+1)}(t)}{n!}(x - t)^n - \frac{M}{n!}(x - t)^n.$$

Setting $t = c$ and using $g'(c) = 0$ gives

$$\frac{(x - c)^n}{n!} \left(f^{(n+1)}(c) - M \right) = 0.$$

Since $c \neq x$, the factor $(x - c)^n$ is non-zero, so $M = f^{(n+1)}(c)$. Substituting this into (*) gives the theorem. $\square$

6 The Remainder Estimation Theorem

Taylor's Theorem is an equality involving an unknown c . Turning it into a usable numerical bound takes one more step.

Theorem 6.1 (Remainder Estimation Theorem). *Suppose there is a constant $M > 0$ with*

$$|f^{(n+1)}(t)| \leq M \quad \text{for all } t \text{ between } a \text{ and } x.$$

Then the remainder in Taylor's Theorem satisfies

$$|R_n(x)| \leq M \frac{|x - a|^{n+1}}{(n+1)!}$$

Proof. Immediate from Theorem 5.2: c lies between a and x , so $|f^{(n+1)}(c)| \leq M$, and $|R_n(x)| = \frac{|f^{(n+1)}(c)|}{(n+1)!} |x - a|^{n+1} \leq \frac{M}{(n+1)!} |x - a|^{n+1}$. $\square$

Example 6.2. Estimate the error if $P_2(x) = 1 - \frac{x^2}{2}$ is used to approximate $\cos x$ at $x = 0.6$.

Solution. Here $f(x) = \cos x$, $a = 0$, $n = 2$, so we need $f^{(3)}(t) = \sin t$, and $|\sin t| \leq 1$ for every t . Take $M = 1$. Then

$$|R_2(0.6)| \leq 1 \cdot \frac{|0.6|^3}{3!} = \frac{0.216}{6} = 0.036.$$

(For comparison, $\cos(0.6) = 0.825336$ and $P_2(0.6) = 0.82$, so the actual error is 0.00534. The bound is honest but generous — it is a guarantee, not a prediction.) $\square$

Example 6.3. Estimate the maximum error when $f(x) = \sqrt{x}$ is approximated by its Taylor polynomial with $n = 2$ about $a = 4$, on the interval $4 \leq x \leq 4.2$.

Solution. We need $f^{(3)}(t) = \frac{3}{8}t^{-5/2}$, which is *decreasing* on $[4, 4.2]$, so its largest value there is at $t = 4$:

$$M = f^{(3)}(4) = \frac{3}{8} \cdot \frac{1}{32} = \frac{3}{256}.$$

The largest value of $|x - 4|$ on the interval is 0.2. Hence

$$|R_2(x)| \leq \frac{M}{3!} (0.2)^3 = \frac{3/256}{6} (0.008) = \frac{0.008}{512} \approx 1.56 \times 10^{-5}.$$

$\square$

Example 6.4. For approximately what values of x may $\sin x$ be replaced by $x - \frac{x^3}{6}$ with an error of magnitude no greater than 4×10^{-3} ?

Solution. Here $P_3(x) = x - \frac{x^3}{6}$ about $a = 0$, so $n = 3$ and $f^{(4)}(t) = \sin t$, giving $M = 1$. Then

$$|R_3(x)| \leq \frac{|x|^4}{4!} = \frac{x^4}{24}.$$

We require $\frac{x^4}{24} \leq 0.004$, i.e. $x^4 \leq 0.096$, i.e.

$$|x| \leq \sqrt[4]{0.096} \approx 0.5566.$$

So for x in $[-0.556, 0.556]$ the cubic approximation is guaranteed accurate to within 4×10^{-3} . $\square$

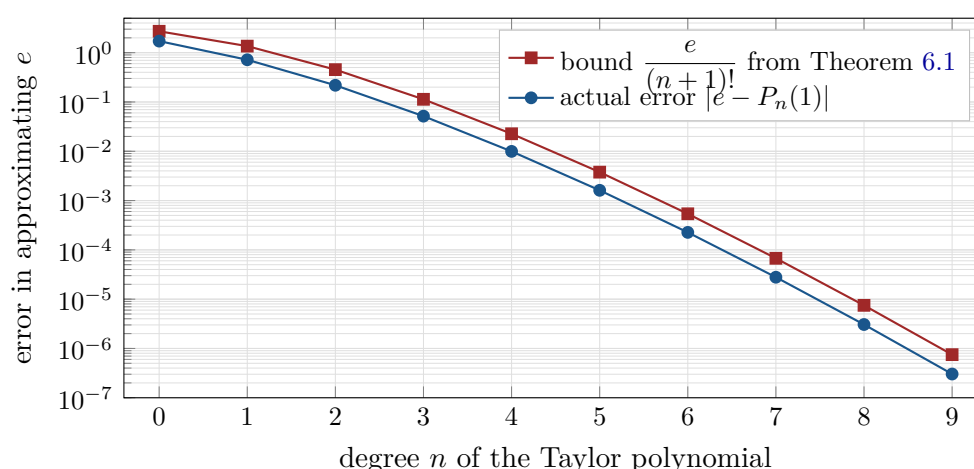


Figure 3: Approximating $e = e^1$ by Maclaurin polynomials. The bound always lies above the true error, as it must, and both fall away like $1/(n+1)!$ — note the logarithmic scale. Nine terms already give seven correct decimals. The factorial in the denominator is what makes Taylor approximation practical.

Try it yourself

Use Theorem 6.1 to find a bound for the error when e^x is replaced by $1 + x + \frac{x^2}{2}$ on the interval $-\frac{5}{6} \leq x \leq \frac{5}{6}$.

Hint: $|f^{(3)}(t)| = e^t \leq e^{5/6} < 3$. *Answer:* $|R_2| \leq \frac{3}{6} \left(\frac{5}{6}\right)^3 \approx 0.289$.

7 Taylor and Maclaurin Series

If f has derivatives of every order, there is no need to stop at degree n .

Definition 7.1 (Taylor series). If f has derivatives of all orders on an interval I containing a , the **Taylor series generated by f at a** is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!} (x-a)^2 + \cdots$$

When $a = 0$ it is called the **Maclaurin series**.

Writing down a series is not the same as it being correct

The definition writes a series for *any* infinitely differentiable f . It does not claim the series adds up to $f(x)$, and in general it need not. Since $f(x) = P_n(x) + R_n(x)$ for every n , the series converges to $f(x)$ at a point x **if and only if**

$$R_n(x) \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

So every claim of the form “ e^x equals its Maclaurin series” is a statement about the remainder, and must be proved. That is what this section does.

Where this connects to the other notes

Convergence of an infinite sum is developed properly in the companion notes *Sequences* and *Series*. Two facts from there are used below, and you may take them on trust for now if you have not met them yet:

- (i) for every fixed real t , $\frac{t^n}{n!} \rightarrow 0$ as $n \rightarrow \infty$ (factorials outgrow exponentials);
- (ii) a power series $\sum c_n(x-a)^n$ converges absolutely inside an interval of radius R about a , the *radius of convergence*.

Theorem 7.2. *The Maclaurin series of e^x converges to e^x for every real x :*

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

Proof. All derivatives of $f(x) = e^x$ are e^x , so $f^{(k)}(0) = 1$ for every k and $P_n(x) = 1 + x + \cdots + \frac{x^n}{n!}$. By Taylor's Theorem,

$$R_n(x) = \frac{e^c}{(n+1)!} x^{n+1}$$

for some c between 0 and x . Since e^t is increasing, e^c lies between $e^0 = 1$ and e^x , so in all cases $e^c \leq \max(1, e^{|x|}) = M$, a constant not depending on n . Hence

$$|R_n(x)| \leq M \frac{|x|^{n+1}}{(n+1)!} \longrightarrow 0$$

as $n \rightarrow \infty$, by fact (i) of the note above. Therefore $P_n(x) \rightarrow e^x$, which is the claim. $\square$

Theorem 7.3. *The Maclaurin series of $\sin x$ converges to $\sin x$ for every real x :*

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

Proof. Every derivative of $\sin x$ is $\pm \sin x$ or $\pm \cos x$, so $|f^{(n+1)}(t)| \leq 1$ for all t and all n . Taking $M = 1$ in Theorem 6.1,

$$|R_n(x)| \leq \frac{|x|^{n+1}}{(n+1)!} \longrightarrow 0$$

for every fixed x . Hence the series converges to $\sin x$ everywhere. $\square$

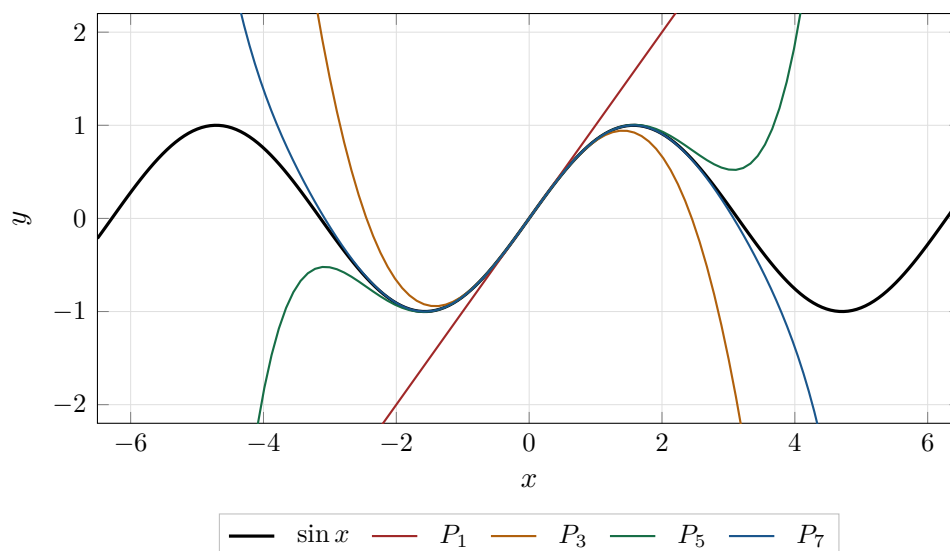


Figure 4: Maclaurin polynomials of $\sin x$. Each one hugs the curve over a longer stretch before peeling away, and the stretch grows without limit as the degree rises — which is the geometric content of “ $R_n(x) \rightarrow 0$ for every x ”. Near $x = 0$ even P_1 is excellent; near $x = 6$ nothing below P_7 is of any use at all.

The same picture in numbers, at $x = 3$, where $\sin 3 = 0.141120$:

	P_1	P_3	P_5	P_7	P_9
value at $x = 3$	3.000000	-1.500000	0.525000	0.091071	0.145312
error	2.859	1.641	0.384	0.050	0.004

The early polynomials are not merely inaccurate, they are wildly wrong — P_3 has the wrong sign. Convergence is a statement about the long run, exactly as it was for sequences.

7.1 The standard series

Series	Valid for
$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \cdots$	all x
$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$	all x
$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$	all x
$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$	$ x < 1$
$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$	$-1 < x \leq 1$
$(1+x)^r = \sum_{n=0}^{\infty} \binom{r}{n} x^n$	$ x < 1$

Setting $x = 1$ in the first series gives a definition of e itself:

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots$$

7.2 A sharper bound for alternating cases

When the Taylor series alternates in sign, the Alternating Series Estimation Theorem (see the note *Series*) usually beats Theorem 6.1: the error is at most the *first term omitted*.

Example 7.4. For what x may $\sin x$ be replaced by $x - \frac{x^3}{3!}$ with error no greater than 3×10^{-4} ?

Solution. The series for $\sin x$ alternates, so truncating after the x^3 term leaves an error no larger than the next term, $\frac{|x|^5}{5!}$. Requiring

$$\frac{|x|^5}{120} < 3 \times 10^{-4} \iff |x|^5 < 0.036 \iff |x| < \sqrt[5]{0.036} \approx 0.514.$$

Moreover the theorem tells us the *direction* of the error: since the first omitted term $\frac{x^5}{120}$ is positive for $x > 0$, the estimate $x - \frac{x^3}{6}$ is an *underestimate* of $\sin x$ for positive x , and an overestimate for negative x — information the remainder bound of Theorem 6.1 cannot give. $\square$

How your calculator really finds $\sin(0.5)$

It adds the Maclaurin series and stops when the next term is too small to matter:

$$0.5 - \frac{0.5^3}{6} + \frac{0.5^5}{120} - \frac{0.5^7}{5040} = 0.5 - 0.0208333 + 0.0002604 - 0.0000016 = 0.4794255.$$

The true value is $0.4794255386\dots$, so four terms give seven correct decimals. The alternating-series bound is what lets the machine know, in advance, how many terms it needs — it does not compute and then check, it checks and then computes.

Convergence of the series is not the same as usefulness

The series for $\sin x$ converges for every x , but at $x = 100$ the terms grow to about 10^{40} before they start to shrink, and rounding error destroys the answer long before convergence sets in. Real software therefore reduces the argument modulo 2π first, and only then uses the series — near 0, where it is fast. Mathematical convergence and numerical practicality are different questions.

Try it yourself

Find the Maclaurin series of $\cos x$ from the definition, and use it to evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$.
Answer: $\frac{1}{2}$.

8 Applications of Taylor's Theorem

Taylor's Theorem is not only a computational device. Because it is an exact equality with a controlled remainder, it proves things.

8.1 Proving inequalities

Example 8.1. Show that $1 - \frac{x^2}{2} \leq \cos x$ for every real x .

Solution. Take $f(x) = \cos x$, $a = 0$ and $n = 2$. Since $f^{(3)}(t) = \sin t$, Taylor's Theorem gives, for some c between 0 and x ,

$$\cos x = 1 - \frac{x^2}{2} + \frac{\sin c}{6} x^3.$$

Case $|x| \leq \pi$. If $x > 0$ then $0 < c < x \leq \pi$, so $\sin c \geq 0$ and $x^3 > 0$, making the remainder ≥ 0 . If $x < 0$ then $-\pi \leq x < c < 0$, so $\sin c \leq 0$ and $x^3 < 0$, and the remainder is again ≥ 0 . Either way $\cos x \geq 1 - \frac{x^2}{2}$.

Case $|x| > \pi$. Then $\frac{x^2}{2} > \frac{\pi^2}{2} > 4$, so $1 - \frac{x^2}{2} < -3 \leq \cos x$, since $\cos x \geq -1$ always. The inequality therefore holds for all $x \in \mathbb{R}$. $\square$

Example 8.2. Show that for every $k \in \mathbb{N}$ and every $x > 0$,

$$x - \frac{x^2}{2} + \cdots - \frac{x^{2k}}{2k} < \ln(1+x) < x - \frac{x^2}{2} + \cdots + \frac{x^{2k+1}}{2k+1}.$$

Solution. By Taylor's Theorem applied to $\ln(1+x)$ at $a = 0$, there is $c \in (0, x)$ with

$$\ln(1+x) = x - \frac{x^2}{2} + \cdots + \frac{(-1)^{n-1}}{n} x^n + \frac{(-1)^n}{n+1} \cdot \frac{x^{n+1}}{(1+c)^{n+1}}.$$

For $x > 0$ the fraction $\frac{x^{n+1}}{(1+c)^{n+1}}$ is positive, so the sign of the remainder is the sign of $(-1)^n$: positive when $n = 2k$ is even, and negative when $n = 2k + 1$ is odd. Taking $n = 2k$ gives the left inequality and $n = 2k + 1$ the right one. $\square$

The idea

Both arguments have the same shape: write the exact Taylor expansion, then argue only about the *sign* of the remainder. No estimate of its size is needed. This is why Taylor's Theorem, with its exact remainder, is so much more powerful than the approximation $f \approx P_n$ on its own.

8.2 A test for local maxima and minima

The usual second-derivative test fails when $f''(a) = 0$. Taylor's Theorem supplies the general answer.

Theorem 8.3. Let $f', f'', \dots, f^{(n)}$ be continuous near a , and suppose

$$f'(a) = f''(a) = \cdots = f^{(n-1)}(a) = 0, \quad f^{(n)}(a) \neq 0.$$

If n is **even**, then f has a local minimum at a when $f^{(n)}(a) > 0$, and a local maximum when $f^{(n)}(a) < 0$. If n is **odd**, a is neither.

Sketch. All the intermediate terms vanish, so Taylor's Theorem reduces to

$$f(x) = f(a) + \frac{f^{(n)}(c)}{n!}(x-a)^n$$

for some c between a and x . If $f^{(n)}(a) > 0$ then by continuity $f^{(n)} > 0$ on a neighbourhood U of a , so for $x \in U$ the sign of $f(x) - f(a)$ is the sign of $(x-a)^n$. When n is even that is positive on both sides, giving a local minimum; when n is odd it changes sign across a , so a is neither a maximum nor a minimum. $\square$

Example 8.4. For $f(x) = x^4$ we have $f'(0) = f''(0) = f'''(0) = 0$ and $f^{(4)}(0) = 24 > 0$. Here $n = 4$ is even and the fourth derivative is positive, so f has a local minimum at 0 — which the second-derivative test could not detect, since $f''(0) = 0$.

8.3 The graph lies above its tangent

Proposition 8.5. *If $f''(x) \geq 0$ on $[a, b]$ (that is, f is convex there), then for every x_0 and x in $[a, b]$,*

$$f(x) \geq f(x_0) + f'(x_0)(x - x_0),$$

i.e. the graph of f lies on or above each of its tangent lines.

Proof. By the Extended Mean Value Theorem (Theorem 4.1) there is c between x_0 and x with

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{(x - x_0)^2}{2}f''(c).$$

The last term is ≥ 0 , since $f''(c) \geq 0$ and $(x - x_0)^2 \geq 0$. $\square$

This is the precise statement behind the remark that a linear approximation *underestimates* a convex function: for $f'' \geq 0$ the tangent lies below the curve everywhere, so $P_1(x) \leq f(x)$. For $f'' \leq 0$ the inequality reverses and the tangent overestimates.

Try it yourself

Use Taylor's Theorem at $a = 0$ with $n = 1$ to prove that $e^x \geq 1 + x$ for every real x .

Hint: the remainder is $\frac{e^c}{2}x^2$, and $e^c > 0$ always. Deduce also, from the Proposition above, that this is the statement that e^x is convex.

9 Summary

- The **linear approximation** $P_1(x) = f(a) + f'(a)(x - a)$ is the tangent line at a ; it is excellent near the base point and deteriorates away from it.
- The **quadratic approximation** adds $\frac{f''(a)}{2}(x - a)^2$, matching curvature as well. The $\frac{1}{2}$ is what makes the formula exact on quadratics.
- Five standard approximations at $a = 0$ cover most cases: $\sin x \approx x$, $\cos x \approx 1 - \frac{x^2}{2}$, $e^x \approx 1 + x + \frac{x^2}{2}$, $\ln(1 + x) \approx x - \frac{x^2}{2}$, $(1 + x)^r \approx 1 + rx + \frac{r(r-1)}{2}x^2$.
- Complicated functions are handled by **combining** standard approximations and discarding all terms above the working order.

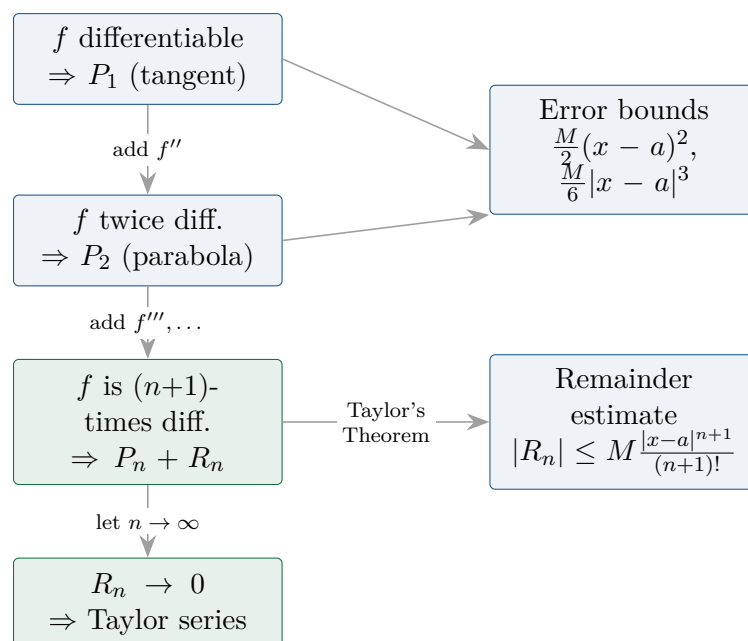


Figure 5: The structure of the chapter. Each row adds one more derivative, and each gives an error bound of the same shape; letting the degree run to infinity turns the polynomial into a series, provided the remainder vanishes.

- **Error bounds:** $|E_1| \leq \frac{M}{2}(x - a)^2$ with $M = \sup |f''|$, and $|E_2| \leq \frac{M}{6}|x - a|^3$ with $M = \sup |f'''|$, the supremum taken over the whole interval between a and x .
- **Taylor's Theorem:** $f(x) = P_n(x) + R_n(x)$ with $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - a)^{n+1}$ for some c between a and x . It contains the Mean Value Theorem as the case $n = 0$.
- **Remainder Estimation Theorem:** if $|f^{(n+1)}| \leq M$ on the interval, then $|R_n(x)| \leq M \frac{|x-a|^{n+1}}{(n+1)!}$.
- The **Taylor series** converges to $f(x)$ exactly when $R_n(x) \rightarrow 0$. This holds for all x for e^x , $\sin x$ and $\cos x$; for $\frac{1}{1-x}$ only on $|x| < 1$.
- For **alternating** series the error is at most the first term omitted, which is both sharper and signed.
- Taylor's Theorem also *proves* results: inequalities such as $1 - \frac{x^2}{2} \leq \cos x$, the general n -th derivative test for local extrema, and the fact that a convex graph lies above its tangents.

Note: After completing these notes, proceed to the [companion Question Bank](#) for self-assessment.

Best wishes.
